Gregory Sinnett

Curriculum Vitae

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Education

2018 **Ph.D., Physical Oceanography**, *Scripps Institution of Oceanography*

2012 **M.S., Physical Oceanography**, *University of Maine*

2004 **B.S., Engineering Physics**, *University of Maine*

Professional History

2025–present **Assistant Project Scientist**, *Scripps Institution of Oceanography*

2023–2025 **Post-Doctoral Researcher**, *Scripps Institution of Oceanography*
Supervisor: Luc Lenain, Ph.D.

2022–2023 **Research Director**, *Walter Munk Foundation for the Oceans*

2021–2022 **Ocean Scientist**, *Scoot Science*

2019–2021 **Post-Doctoral Researcher**, *University of California, Irvine*
Supervisor: Kristen Davis, Ph.D.

2018–2019 **Post-Doctoral Researcher**, *Scripps Institution of Oceanography*
Supervisor: Falk Feddersen, Ph.D.

2012–2018 **Graduate Research Assistant**, *Scripps Institution of Oceanography*
Supervisor: Falk Feddersen, Ph.D.

2010–2012 **Graduate Research Assistant**, *University of Maine*
Supervisor: Neal Pettigrew, Ph.D.

2004–2010 **Technology Development Engineer**, *Fairchild Semiconductor*

Funding Received

2026–2029 **California Alert and Risk Mapping (C-ALARM) Suite: Next-Generation C-HARM Forecasts and Automated Alerts of Domoic Acid Risk**, *OPC*, $1.5M
w/ C. Anderson, L. Lenain (SIO), R. Kudela, D. Robinson (UCSC), E. Bjorkstedt (Cal Poly Humboldt), A. Pasulka (Cal Poly SLO).

Publications

Journal Publications

2025 **Sinnett, G.**, Lenain, L., Braham, E., Chaudhry, N. A., & Dinasquet, J. Contribution of large marine aerosols in phytoplankton dispersal. *Environmental Science & Technology*, **59**(14), 7348–7356. doi:10.1021/acs.est.4c14473

Updated: November 2026

2022 **Sinnett, G.**, Ramp, S. R., Yang, Y. J., Chang, M., Jan, S., & Davis, K. A. Large amplitude internal wave transformation into shallow water. *Journal of Physical Oceanography*, **52**, 2539–2554. doi:10.1175/JPO-D-21-0273.1

2022 Ramp, S. R., Yang, Y.-J., Jan, S., Chang, M.-H., Davis, K. A., **Sinnett, G.**, et al. Solitary waves impinging on an isolated tropical reef: Arrival patterns and wave transformation under shoaling. *Journal of Geophysical Research: Oceans*, **127**, e2021JC017781. doi:10.1029/2021JC017781

2020 **Sinnett, G.**, Davis, K. A., Lucas, A. J., Giddings, S. N., Reid, E., Harvey, M. E., & Stokes, I. Distributed temperature sensing for oceanographic applications. *Journal of Atmospheric and Oceanic Technology*, **37**, 1987–1997. doi:10.1175/JTECH-D-20-0066.1

2019 **Sinnett, G.**, & Feddersen, F. The nearshore heat budget: Effects of stratification and surfzone dynamics. *Journal of Geophysical Research: Oceans*, **124**, 8219–8240. doi:10.1029/2019JC015494

2018 **Sinnett, G.**, & Feddersen, F. The competing effects of breaking waves on surfzone heat fluxes: Albedo versus wave heating. *Journal of Geophysical Research: Oceans*, **123**, 7172–7184. doi:10.1029/2018JC014284

2018 **Sinnett, G.**, Feddersen, F., Lucas, A. J., Pawlak, G., & Terrill, E. Observations of nonlinear internal wave run-up to the surfzone. *Journal of Physical Oceanography*, **48**, 531–554. doi:10.1175/JPO-D-17-0210.1

2016 **Sinnett, G.**, & Feddersen, F. Observations and parameterizations of surfzone albedo. *Methods in Oceanography*, **17**, 319–334. doi:10.1016/j.mio.2016.07.001

2014 **Sinnett, G.**, & Feddersen, F. The surf zone heat budget: The effect of wave heating. *Geophysical Research Letters*, **41**. doi:10.1002/2014GL061398

Proceedings and Technical Reports

2018 **Sinnett, G.** The nearshore heat budget. Ph.D. dissertation, University of California, San Diego. ProQuest

2016 **Sinnett, G.**, Feddersen, F., Lucas, A. J., Pawlak, G., & Terrill, E. Non-linear internal waves pulse cold water into the shallow inner-shelf and surfzone. *VIIIth International Symposium on Stratified Flows*. PDF

2012 **Sinnett, G.** Circulation and transport in Casco Bay, Maine. M.S. thesis, University of Maine, Orono, Maine.

Presentations

2024 "Coastal flood risk assessment along data-sparse coastlines" (invited). Flood Hazards and Installation Resilience Workshop, Annapolis, MD, August 11–12.

2023 "Distributed Temperature Sensing for Oceanographic Applications." Oceanology International, San Diego, CA, February.

2020 "Global Ocean Warming" (invited). American Geographical Society, November.

2020 "Distributed Temperature Sensing (DTS) Application to Oceanography." Ocean Sciences Meeting, San Diego, CA, February.

2019 "Distributed Temperature Sensing (DTS) Application for Nearshore Internal Wave and Heat Budget Observations" (invited). Scripps Institution of Oceanography, November.

2018 "A Detailed Nearshore Heat Budget." Eastern Pacific Ocean Conference, Timberline Lodge, OR, September.

2018 "A Detailed Nearshore Heat Budget," Abstract CD12A. Ocean Sciences Meeting, Portland, OR, February 12–16.

2017 "Observations of Nonlinear Internal Wave Runup into the Surfzone." Gordon Research Conference on Coastal Ocean Dynamics, Maine.

2016 "Observations of Non-Linear Internal Wave Run-up into the Surfzone." AGU Fall Meeting, San Francisco, CA, December.

2016 "Observations of Non-Linear Internal Waves Pulsing Cold Water to the Surfzone." Eastern Pacific Ocean Conference, Timberline Lodge, OR, September.

2016 "Non-Linear Internal Waves Pulse Cold Water into the Shallow Inner-Shelf and Surfzone." VIIIth International Symposium on Stratified Flows, San Diego, CA, August.

2016 "Surging Non-Linear Internal Waves Deliver Cold Inner-Shelf Water to the Surfzone." Ocean Sciences Meeting, New Orleans, LA, February.

2014 "Observations of Surfzone Albedo." AGU Fall Meeting, San Francisco, CA, December.

2014 "Characterizing Heat Content and Spatio-Temporal Variability of Temperature in the Surf Zone." Ocean Sciences Meeting, Honolulu, HI, February.

2014 "Ocean Currents for 9th Grade." NSF GK-12 Meeting, La Jolla, CA, May.

Public Presentations

2023 "Exploring El Niño" (invited). La Jolla Public Library, La Jolla, CA, April.

2022 "Beneath La Jolla's Shores" (invited). La Jolla Public Library, La Jolla, CA, April.

2016 "Physical Oceanography Near the La Jolla Coast" (invited). La Jolla Public Library, La Jolla, CA, April.

Selected Press

2015 *EOS* Research Spotlight (5 February): "Wave energy affects the surf zone heat budget." doi:10.1029/2015EO023167

2015 *Geophysical Research Letters* editor highlight (16 January): "Wave heating effects on the surf zone heat budget." Link

2014 *Nature News* highlight (15 October): "Surf zones warmed from within." doi:10.1038/nature.2014.16148

Field Experience

2024–2026 **Flood Forecasting and Prediction**, *Various locations*

Coordinated and oversaw fieldwork, data analysis and product development during a multi-year and broad effort to map and develop flood products at Navy installations around the world. Observations from aircraft, including lidar, visual, and hyperspectral imagery were collected over a wide range of surface environments. Pluvial, fluvial and coastal inundation flood products were developed from these datasets for a 100 year flood scenario.

2024 **Large Aerosol Observations**, *Scripps Pier, CA*

Collaborated with biological oceanographers at Scripps to deploy a suite of aerosol observing equipment to quantify large ocean aerosols (up to 90 μm). Results of this two-month study confirmed higher levels of large aerosols than predicted by standard wind-driven parameterizations and demonstrated, with direct evidence, large aerosols regularly transport intact ocean phytoplankton cells.

2023 **Alpine Lake Climate Observatory**, *Lake Altaussee, Austria*

Developed and funded a long-duration observatory at a remote and pristine Alpine lake to study climate effects on sensitive lake ecological systems. Coordinated research effort among three international institutions and lead PI, installing a research-grade weather station, mooring to monitor temperature and oxygen, and sensors to record lake level and outflow temperature. Experimental goals: quantify seasonal and climate scale inputs to stratification, lake temperature fluctuation and relation to fish populations. This project supports the Austrian Federal Ministry of Water Management climate monitoring goals.

2019 **Dongsha Internal Waves**, *Dongsha, Taiwan*

Coordinated and executed a two-month multi-institutional field campaign in the remote Dongsha Atoll (South China Sea) to deploy and recover over 160 instruments including 8 moorings, a DTS cable system, two wire-walkers, bottom temperature sensors, and several ADCPs. Experimental goals: characterize the internal wave effects on the fringing fore-reef system, observe and quantify large amplitude internal wave shoaling and breaking mechanisms, and isolate factors contributing to lagoon circulation, exchange and occasional hypoxia.

2018 **DTS Internal Wave Experiment**, *Scripps Beach, CA*

Characterized a Distributed Temperature Sensing (DTS) fiber optic cable system for novel use in nearshore physical oceanography applications. Deployed three $\sim$2 km fiber optic cables in precise locations to both experimentally test the new sensing platform and observe the internal wave field onshore of the La Jolla canyon system.

2017 **Inner-Shelf Dynamics Experiment**, *Point Sal, CA*

Eight days aboard the R/V *Sally Ann* conducting nearshore drifter releases and survey transects. This coordinated experiment across many institutions was designed to develop and improve the predictive capability of a range of numerical models, simulate circulation, density, and the surface wave field across the inner shelf associated with a broad array of physical processes and complex bathymetry.

2015 **CSIDE Experiment**, *Imperial Beach, CA*

Multiple surfzone dye releases tracked with a variety of underwater, surface and airborne instruments. Provided diving, underway CTD, jet ski operations and shore support.

2014–2015 **SIO14 Experiment**, *Scripps Beach, CA*

Designed, deployed and maintained an array of over 60 instruments in water 0–18 m for nine months to characterize new terms in the nearshore heat budget related to previously ignored surfzone dynamics.

2012 **SIO12 Experiment**, *Scripps Beach, CA*

Recovered and analyzed data from 8 temperature sensors in water 0–7 m as well as numerous meteorological equipment to study the nearshore heat budget and wave heating effect in the surfzone.

Research Voyages

Various (Altaussee): CTD casts, mooring deployment.

R/V Atoll 6: CTD and microCTD casts, mooring deployment, DTS deployment/recovery, wirewalker deployment/recovery, dives.

R/V Sally Ann: Towed ADCP, CTD, drifter deployment, dives, moored instrument deployment/recovery.

R/V New Horizons: Mooring recovery/redeployment, CTD stations.

R/V Ocean Starr: Mooring recovery/redeployment, CTD stations.

R/V Connecticut: Mooring recovery/redeployment.

Diving — AAUS certified science and rescue diver with ∼150 scientific dives.

Research Related Service

2016–present — **Journal Peer Reviewer**: *Journal of Physical Oceanography*, *Journal of Geophysical Research: Oceans*, *Oceanography*, *Methods in Oceanography*, *Journal of Oceanography*.

Proposal Reviewer: National Science Foundation, Physical Oceanography.

Teaching, Mentoring and Volunteering

2025–2026 — **Masters Thesis Committee member**: Sarina Ghadiali.

2023–present — **Student advisor**, Scripps Institution of Oceanography.

2023–2024 — **Oceanography Advisor**, Trofix.

2020–present — **Visiting Scientist**: Spreckels Elementary, The Children's School, High Tech High.

2018–present — **Alumni Mentor**, University of Maine.

2013–2018 — **Volunteer Scientist**, SCOPE (Scripps Community Outreach for Public Education).

2013–2014 — **NSF GK-12 Embedded Instructor**, Kearny High School.